

# Chasing Thirty Million Words

Parental Speech, Gender, and Early Child Development in Timor-Leste

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## The paper in one slide

**Q.** Can a low-cost intervention about child-directed speech (CDS) raise early child development?

**Design.** Village-level RCT, Timor-Leste: mother / father / control; info + video + calendar + picture book.

**Result.** Child vocabulary  $+0.22 / 0.32$  SD — **only via mothers**; fathers only in matrilineal/matrilocal villages.

Ambitious, valuable paper. Two comments: (1) what is the mechanism? (2) what is the contribution?

## Point 1: an effect in search of a mechanism

The intervention's implied chain — *info* → *beliefs* → *more talk* → *child language* — has no support in its middle links. Every candidate *channel* is flat:

| Candidate channel                               | Moves? |
|-------------------------------------------------|--------|
| <i>Perceived</i> returns / cost / self-efficacy | No     |
| Self-reported CDS                               | No     |
| Recorded speech to / by the child               | No     |

Yet the *result* — *child vocabulary* +0.22/0.32 SD — moves. The outcome rises while every channel stays flat.

## Point 1: so what moved?

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- The only *measured* movers are **book possession** and **video recall**. **Two roads** to pin down the mechanism — both with **data you already have**:
- **Road A — materials.**
  - (1) *Teaching to the test?* Split items **book vs non-book** — is the gain only on items the book depicts?
  - (2) *Really the book?* Split **Table 8** (materials found / used) by kinship — is the book used more where the effect appears?
- **Road B — speech mismeasured.** All-day vocalization and conversational-turn counts may be imprecise to capture CDS. Measure it where it lives:
  - (1) only when the *parent is present* — empty/absent hours aren't “low talk,” drop them;
  - (2) *strip TV/radio*, keep human voice;
  - (3) *distinct words the parent says to the child while talking* — transcribe the recordings (they have them; VTC gives turns, not words).

## Point 2: what's the contribution?

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- **If the book is the active ingredient: it's a *tool*, not belief-change.**
  - **Dupas et al. (2025, Ghana):** info + reminders moved beliefs & self-reports — but objective child outcomes stayed null.
  - **You:** no belief change — yet child vocabulary rises.
- **The gender angle (*mechanism-independent*): which parent to target.**
  - **Jeong et al. 2024, Luoto et al. 2021:** additive (fathers on top of mothers) — adding fathers improves nothing; can't ask “father instead of mother.”
  - **You:** a substitution design (father vs mother) and the kinship / bargaining-power moderator. Answer: mothers; fathers only where women hold power.

## More comments

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1. **Matriliney isn't randomized** — villages may differ on other dimensions (Lowes 2022).  
*Control for co-varying traits.*
2. **Number of children** — in larger families parental *time* per child is binding (CDS may not rise), yet the book can *spill over* to siblings — and you already record a **second child**.  
*Heterogeneity by family size; exploit the second-child recording.*
3. **Numeracy emerges *later* than words** — counting / knowing your age develops years after early vocabulary. Measured at the 1-yr endline (children ~2–5 yo), most are too young, so the numeracy null is likely *premature*, not a true zero. *Use the unused 2<sup>nd</sup> endline — or set a higher age floor for numeracy than the >30 mo used for verbal.*

**Congratulations on the paper — and thank you.**